MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG5DSCMAT301-LIMITS AND CONVERGENCE

Duration: 2 hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability

Part A
Answer any five questions. Each questions carries 2 marks

1. Define the tail of a sequence. Give an example to illustrate the definition. [U] [1]
2. State Monotone Convergence Theorem [K] [2]
3. Discuss the convergence of the sequence $\left(1, \frac{1}{2}, 3, \frac{1}{4}, \ldots.\right)$ [An][4]
4. When do you say that a sequence (x_n) is properly divergent? [K] [5]
5. Apply the Comparison Test to determine whether the series $\sum_{n=1}^{\infty}\left(\frac{1}{n^2+n}\right)$ converges or diverges. [A] [5]
6. State Raabe's Test for determining the convergence or divergence of a series. [K] [6]
7. Explain sequential criterion for limits of functions [U] [8]
8. Describe when a function $f(x)$ is said to be bounded in a neighbourhood of a cluster point. [U] [8]

Part B
Answer any five questions. Each questions carries 6 marks

9. Apply the definition of the limit of a sequence to show that a sequence in R can have at most one limit. [A] [1]
10. Prove that every convergent sequence of real numbers is bounded. [An][1]
11. State and prove Monotone Subsequence Theorem. [An][3]
12. Explain the nature of the sequence $\left(\frac{1}{1}+\frac{1}{2}+\cdots+\frac{1}{n}\right)$. [U] [4]
13. Determine the nature of the series whose nth term is $\frac{n!}{3.5.7\ldots.(2n+1)}$. [A] [6]
14. Apply Dirichlet's Test to determine the convergence of $\sum_{n=1}^{\infty}(a_n \sin nx)$, where (a_n) is decreasing and converges to zero. [A] [7]
15. Discuss why $\lim_{x\to 0}\left(x \sin\left(\frac{1}{x}\right)\right)$ approaches 0 as $x \to 0$ [U] [8]
16. Describe, using the definition of limit at infinity, why $\lim_{x\to+\infty} x^n = +\infty, n \in N$. [U] [8]

Part C
Answer any three questions. Each questions carries 10 marks

17. State and prove squeeze theorem for sequences and hence find $\lim_{n\to\infty}\left(\frac{\sin n}{n}\right)$. [An][1]

18. a) Show that every convergent sequence is a Cauchy sequence. Is the converse true? Explain your answer.
    b) Show that a sequence of real numbers is convergent if and only if it is a Cauchy sequence. [An][4]

19. Let (x_n) and (y_n) be two sequences of positive real numbers and suppose that for some $L > 0$, $\lim(\frac{x_n}{y_n}) = L$. Apply the limit properties to show that $\lim(x_n) = +\infty$ *if and only if* $\lim(y_n) = +\infty$. [A] [5]

20. a) Define the rearrangement of a series and state rearrangement theorem on the absolute convergence.
    b) Show that if a series $\sum x_n$ is convergent, then any series obtained from it by grouping the terms is also convergent and to the same limit. [A] [6]

21. a) Explain $i)$ a cluster point of a subset of R, $ii)$ limit of a function at a cluster point.
    b) Describe how a cluster point of a subset of R can be characterized using a sequence in the subset. [U] [8]

22. a) Interpret whether the one-sided limits of the function $g(x) = e^{\frac{1}{x}}$, at $x = 0$ exists or not.
    b) Explain whether the function $f(x) = sgn(x)$ has distinct right-hand and left-hand limits at $x = 0$. [U] [8]